

Prompt Evaluation

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

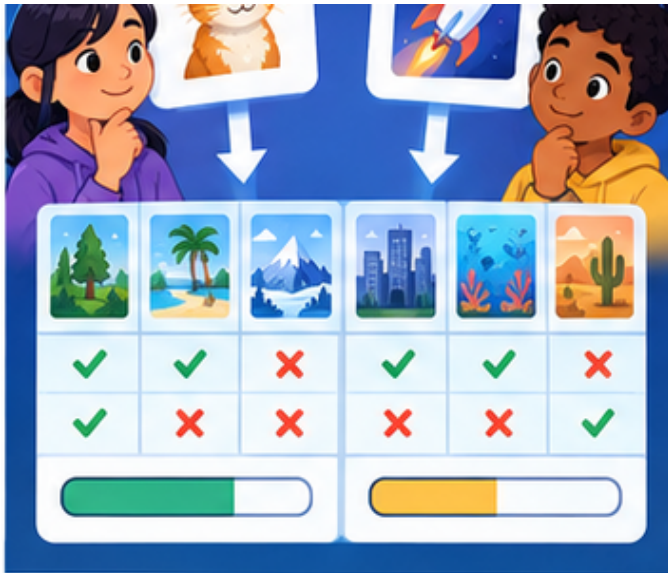
Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 9: Prompt Evaluation

Prompt claims require a fixed test set and repeatable scoring.



CREATOR MISSION

A/B test two prompts across six cases.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Structured outputs and validation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

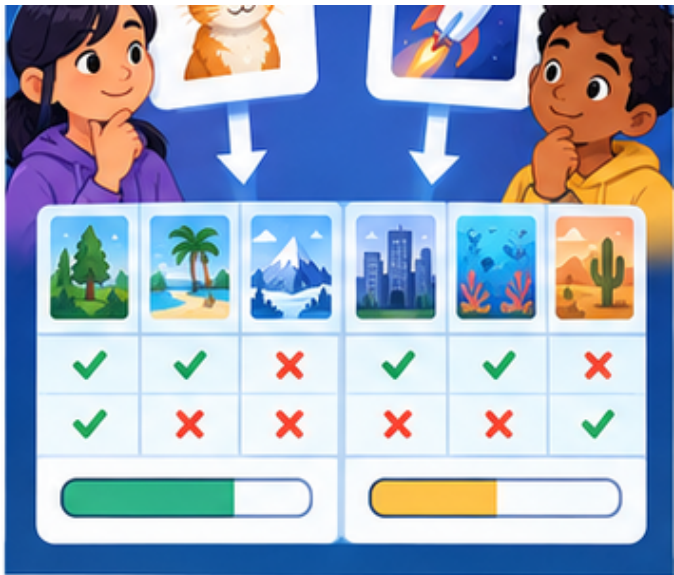
By the end, I can evaluate prompts with paired cases and a predeclared mandatory criterion.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use prompt evaluation to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings.

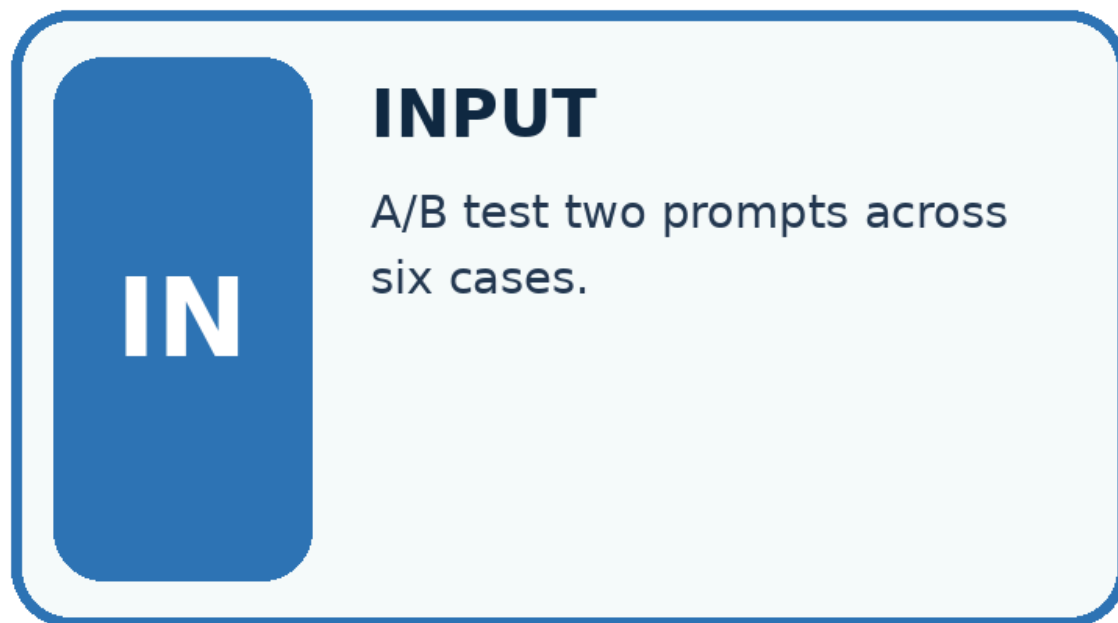
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Prompt claims require a fixed test set and repeatable scoring.



OUT

EVIDENCE

A/B test two prompts across six cases.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Controlled prompt experiments — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Check or calculation: Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.

Conclusion and limit: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Remove the gate and retain total passes as the only criterion.

Prediction: The record is a tie; another predeclared tie policy or further development evidence is needed.

Reason: Both prompts have four passes on six questions.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

A/B TEST

compare two versions under
the same cases

2

TEST SET

fixed examples used for
evaluation

3

SCORE

measured rubric result

4

VARIANCE

results differ across runs or cases

Word · My own example · Not this / boundary

Word: A/B test

My own example: A comparison of two versions using a planned assignment of comparable cases and the same outcome measure.

Not this / boundary: _____

Word: test set

My own example: Data reserved for evaluating a selected system; final test data must not influence its development.

Not this / boundary: _____

Word: score

My own example: A numerical value assigned using a specified rule; it is not automatically a probability.

Not this / boundary: _____

Word: variance

My own example: A measure of spread around a mean; informally, variation across repeated results. Specify which meaning is intended.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Paired case

Meaning: The same case evaluated under each compared procedure

Word: Rubric

Meaning: Prespecified criteria for judging an output

Word: Regression

Meaning: A previously passing condition that fails after a change

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 9 • WORKED EVIDENCE

- 1 Prompt A scores 20/30 across six cases.
- 2 Prompt B scores 25/30 on the same cases and rubric.
- 3 B improves evidence but loses one clarity point on long passages.
- 4 Repeat runs and inspect per-case failures before choosing.

1. Prompt A scores 20/30 across six cases.

2. Prompt B scores 25/30 on the same cases and rubric.

3. B improves evidence but loses one clarity point on long passages.

4. Repeat runs and inspect per-case failures before choosing.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Evaluation

A fictional Study Buddy compares two prompts across six development questions.

The facts we will use

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common rubric.

Same total, different mandatory-case results

	Total passes /6	Refusal passes /2
Prompt A	4	0
Prompt B	4	2

Read a cell using both its row and column labels.

Prompt Evaluation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired total

A and B both pass 4/6.



2. Different failures

A alone passes 3,4; B alone passes 5,6.



3. Mandatory slice

A passes 0/2 refusal cases; B passes 2/2.



4. Selection

B clears the declared gate, but its failures 3,4 remain visible.

CHANGE ONE THING

Remove the gate and retain total passes as the only criterion.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Paired total

What happens and why: A and B both pass 4/6.

Step: 2. Different failures

What happens and why: A alone passes 3,4; B alone passes 5,6.

Step: 3. Mandatory slice

What happens and why: A passes 0/2 refusal cases; B passes 2/2.

Step: 4. Selection

What happens and why: B clears the declared gate, but its failures 3,4 remain visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result. Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

What this example does not establish

Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A single impressive output proves one prompt is best. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Prompt selection on development cases cannot be reported as independent final performance.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is B’s selection defensible here without calling it universally superior?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Freeze six cases, run A and B with one changed variable, score blind with the same rubric, and explain wins and regressions.

Freeze six cases, run A and B with one changed variable, score blind with the same rubric, and explain wins and regressions.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what

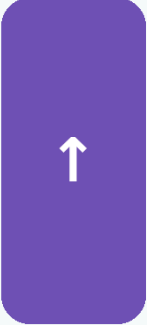
did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

THIS WEEK’S CHECKPOINT Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

A/B test two prompts across six cases.

CHALLENGE A/B test two prompts across six cases.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

On five fixed questions, A passes four and B passes three; B looks more attractive.

The question

What conclusion is supported?

Worked explanation

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Evaluate prompts on the same questions

Evidence or stage · Value or result

Evidence or stage: Prompt A

Value or result: 4/5

Evidence or stage: Prompt B

Value or result: 3/5

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

14. Try a new case independently

Independent case: Changed independent case

Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A source check is mandatory. Which can be accepted as-is?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

On four shared tasks, A scores 1,1,0,1 and B scores 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.